

CASE STUDY

Assortment Rationalization for a Leading European Retailer

What to retain, replace and eliminate across 5,000 grocery products to lift profitability, release working capital and improve the shopper experience

Analysis of 5,000 products · 6 categories · 5 brands · Figures in EUR

EXECUTIVE SUMMARY

One-third of the product range is underperforming and dilutes the value generated by the remaining

€454.7M

Annual sales across 5,000 products; aggregate gross margin €114.7M (25.2%)

37.3%

of products (1,866) sell below cost and erase **€83.1M** of margin

39.3%

of products are Stars that deliver 64.7% of sales and €170.0M margin

53.6%

margin rate on the remaining range once the 1,873 flagged products are removed

WHAT WE FOUND

Losses are spread evenly across every category and brand, so this is a product-level problem, not a category one. Customer ratings do not predict sales, and shelf space is allocated with no link to what a product sells.

WHAT WE RECOMMEND

Delist 1,653 loss-making, slow or unloved SKUs; reprice or replace 220 well-loved loss-makers; give the freed space (~33% of shelf) to 1,186 Stars and efficient sellers; promote 298 high-rated, high-margin products that are under-selling.

THE BRIEF

Too many products, not enough of them pulling their weight

Problem

The retailer offers a wide grocery range, but many items contribute little to revenue or demand. Carrying all of them drives up inventory, handling and space costs.

Objectives

- 1. Identify underperforming items
- 2. Find high-demand items that need more space or stock
- 3. Recommend rationalization moves that raise profit, cut working capital and lift satisfaction

Data and approach

5,000 products × 15 fields, no missing values: price, cost, units sold, SKU count, shelf space, customer rating, purchase frequency, sales and gross margin.

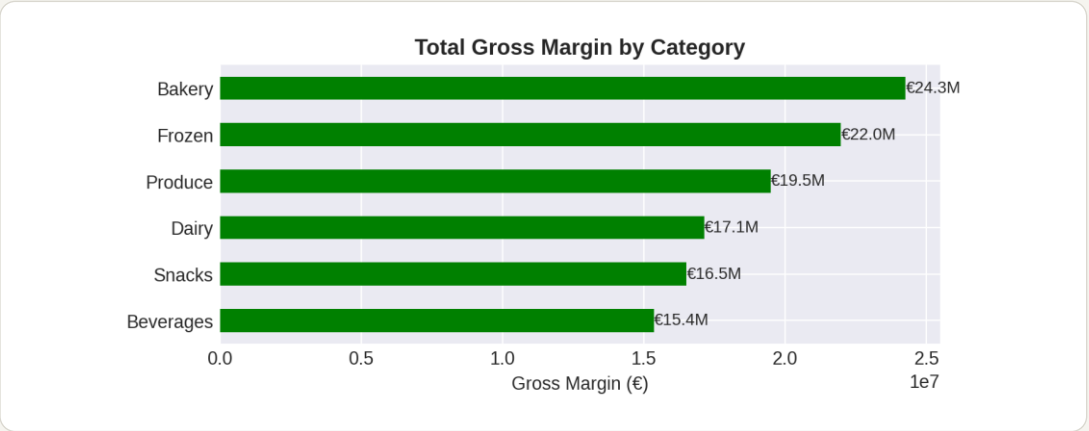
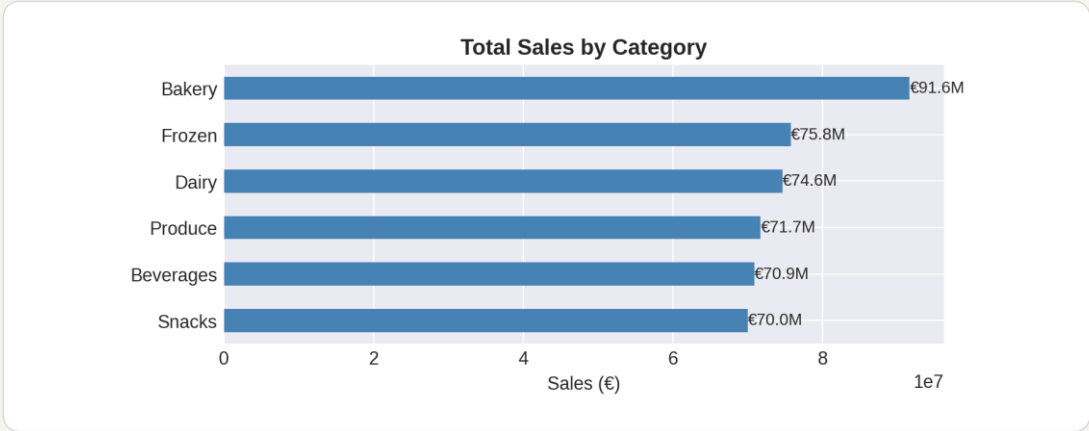
#	Analysis
1	Descriptive analytics: category, brand, top and bottom
2	Customer demand: ratings and purchase frequency
3	Inventory: stock turnover and shelf productivity
4	Profitability: margin segmentation
5	Recommendations and revised assortment

SECTION 01

Descriptive analytics

Sales and gross margin by category and brand, and who the top and bottom performers are.

Bakery leads on size; Frozen converts sales to margin best



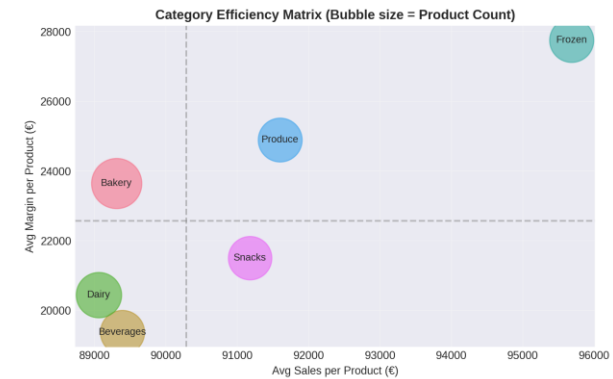
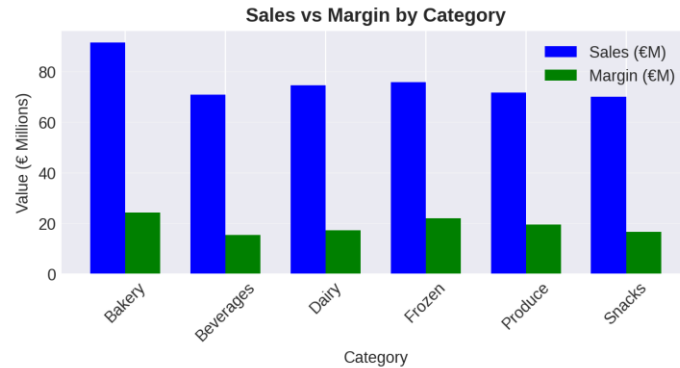
FINDING

Bakery is the largest category at €91.6M sales (20.2%) and €24.3M margin. The other five sit in a tight €70–76M band. Frozen is 2nd on both sales and margin (€22.0M); Dairy drops from 3rd to 4th on margin and Beverages is last (€15.4M).

ANALYTICAL INSIGHT

Category sales are balanced, so no category should be cut wholesale. The margin spread is the story: Frozen earns a 29.0% margin rate vs 21.7% for Beverages. Beverages and Dairy are where cost and price terms need attention first.

Frozen and Produce earn more per product than the rest



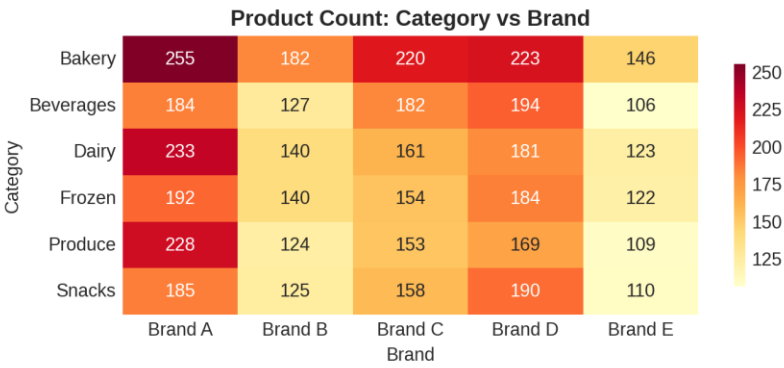
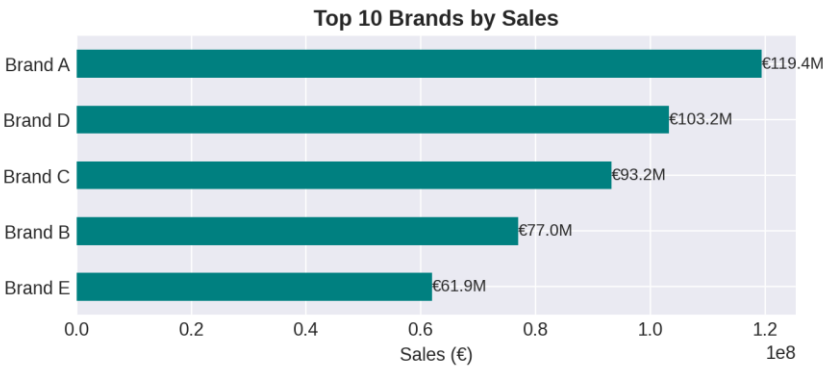
FINDING

Margin is a fraction of sales in every category. Per product, Frozen leads (€95.7k sales, €27.7k margin) and Produce follows. Bakery has the most products (1,026) but average sales per product; Dairy and Beverages sit bottom-left on both axes.

ANALYTICAL INSIGHT

Bakery's lead comes from breadth, not productivity, which makes it the first candidate for SKU trimming. Dairy and Beverages combine below-median sales and margin per product, so they need both range cuts and better terms. Frozen is the model to protect and extend.

Brand A leads in sales, range and margin rate



FINDING

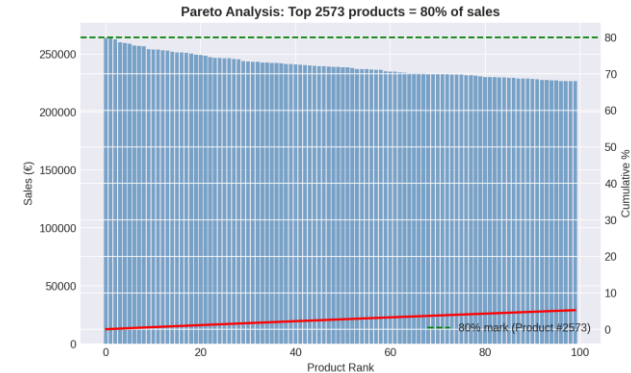
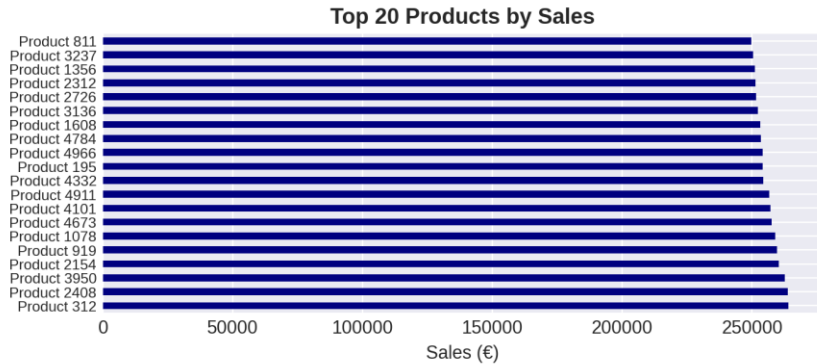
Brand A sells €119.4M from 1,277 products and has the best margin rate (28.5%). Brand D is 2nd on sales (€103.2M) but earns only 22.9%. Brand E is smallest (€61.9M) and weakest on margin (20.7%). Brand A has the widest range in 4 of 6 categories.

ANALYTICAL INSIGHT

Brand D is the key negotiation target: high volume at a below-average margin. Brands D and E both have 38.8% of their products at a negative margin. With 100+ products for every brand in every category, the shelf holds a lot of duplication that can be thinned out.

DESCRIPTIVE ANALYTICS · TOP PERFORMERS

Top sellers each bring ~€250k, but sales are spread thinly



FINDING

The top 20 products each sell about €250–264k (e.g. Product 312 Dairy, 2408 Produce, 3950 Snacks). There is no classic 80/20 curve: it takes 2,573 products (51%) to reach 80% of sales.

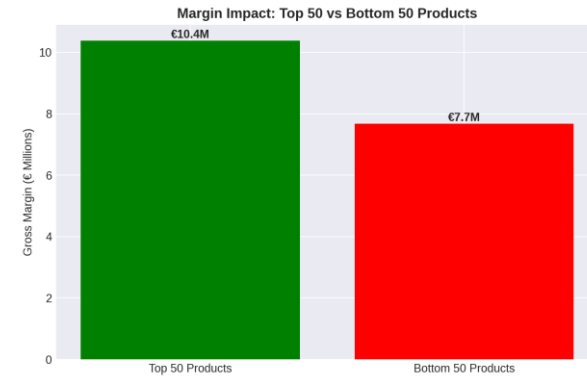
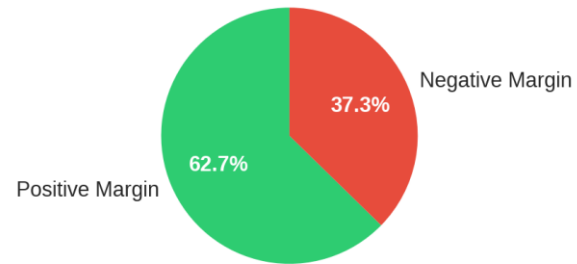
ANALYTICAL INSIGHT

Because sales are spread so evenly, cutting by sales rank alone would remove real revenue. Rationalization has to be driven by profit, turnover and shelf productivity. Top sellers are not all equal: Product 919 sells €259.6k but earns only €64.3k margin.

DESCRIPTIVE ANALYTICS · BOTTOM PERFORMERS

1,866 products sell at a loss and wipe out €83M of margin

Product Distribution by Margin Status



FINDING

37.3% of products (1,866) have a negative gross margin. Together they account for 18.9% of sales but lose €83.1M, 42% of the €197.8M earned by profitable items. The bottom 50 alone lose €7.7M, erasing 74% of what the top 50 earn (€10.4M).

ANALYTICAL INSIGHT

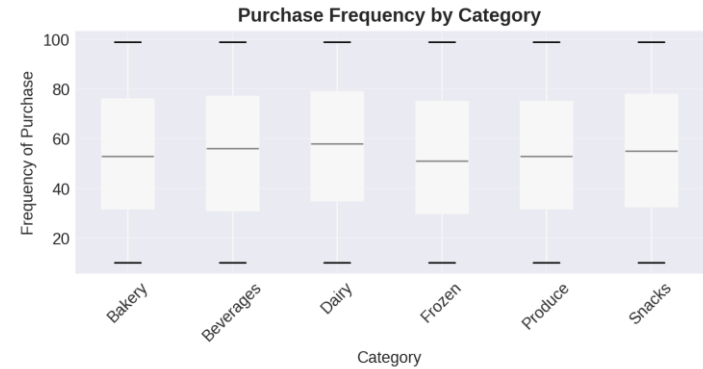
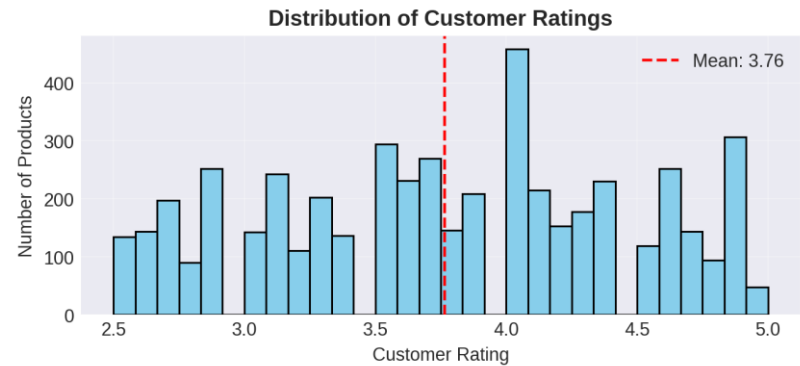
The worst items lose far more than they sell (Product 4998: €11.1k sales, –€185.4k margin), which means cost sits above price. Every one of these products needs a decision: reprice, renegotiate cost or delist. Bottom-10 sellers by revenue are all loss-making too.

SECTION 02

Customer demand analysis

Customer ratings and purchase frequency, and whether they explain what sells.

Ratings and purchase frequency look the same in every category



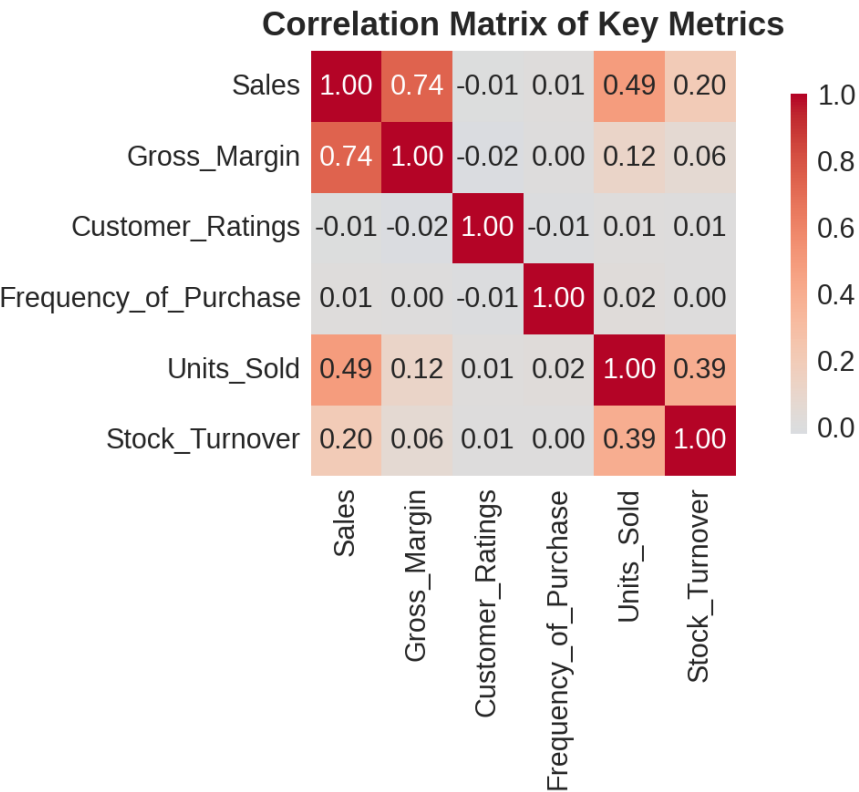
FINDING

Ratings span 2.5–5.0 with a mean of 3.76; 25% of products score 4.3+. Category averages fall in a tight 3.74 (Dairy) to 3.80 (Snacks) band. Purchase frequency has a median of 54 in every category (Dairy highest at 58). 1,313 products are in the top frequency quartile.

ANALYTICAL INSIGHT

No category is loved or ignored by shoppers as a whole, so demand has to be judged product by product. The popular set to protect is the 964 products rated 4.5+, especially where frequency is also in the top quartile.

Ratings and frequency do not predict sales or margin



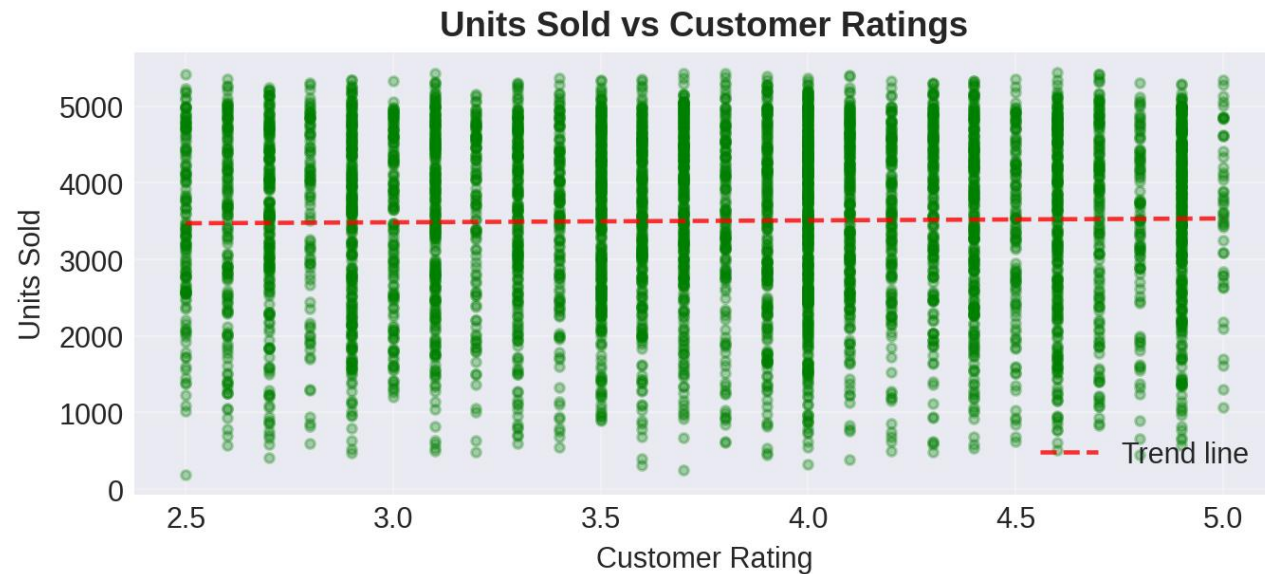
FINDING

Customer rating correlates with sales at -0.01 , with frequency at -0.01 and with margin at -0.02 . Frequency correlates with sales at just 0.01 . Sales track margin (0.74) and units sold (0.49).

ANALYTICAL INSIGHT

Demand signals and financial results are unrelated in this range. So a high rating cannot be used to defend a loss-maker, and a low rating alone is not a reason to delist. Ratings are best used as a tie-breaker between otherwise similar products.

Well-rated products sell no more units than poorly rated ones



FINDING

The trend line is flat: products rated 2.5 and 5.0 both sell about 3,500 units. 1,107 products are rated 4.0+ yet sit below median sales.

ANALYTICAL INSIGHT

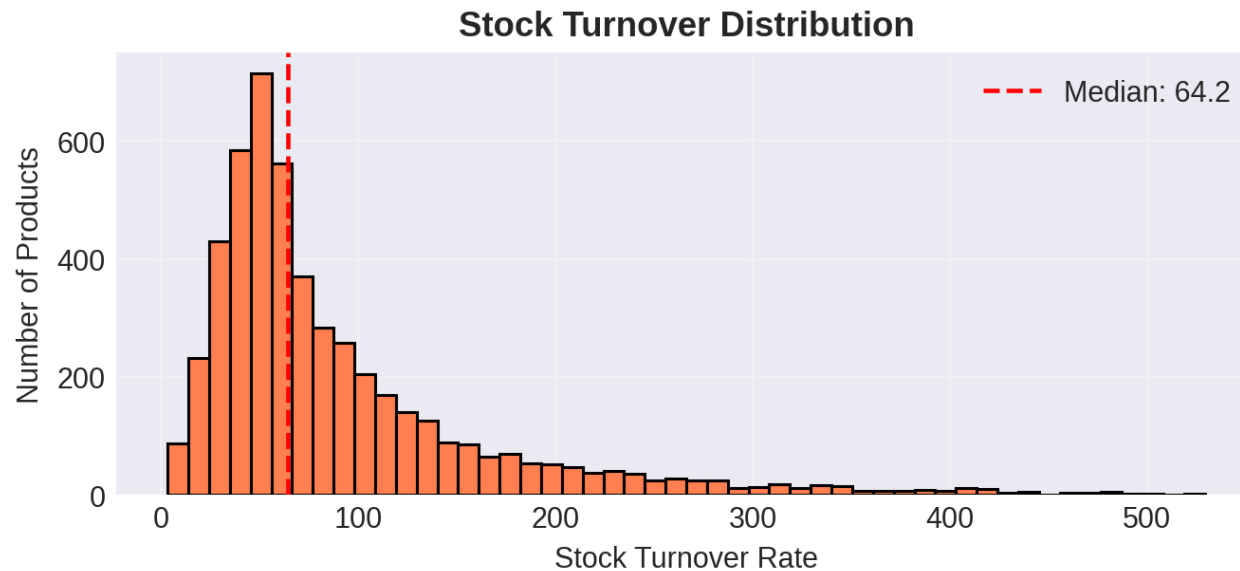
Many shoppers like products they rarely find or notice. That points to a visibility and availability gap, not a demand gap. These well-rated, under-selling products are the growth pool (348 of them also clear a 20% margin) for better placement and promotion.

SECTION 03

Inventory optimization

Stock turnover to find slow movers, and shelf-space productivity to find over- and under-used space.

A quarter of the range turns at less than half the speed of the rest



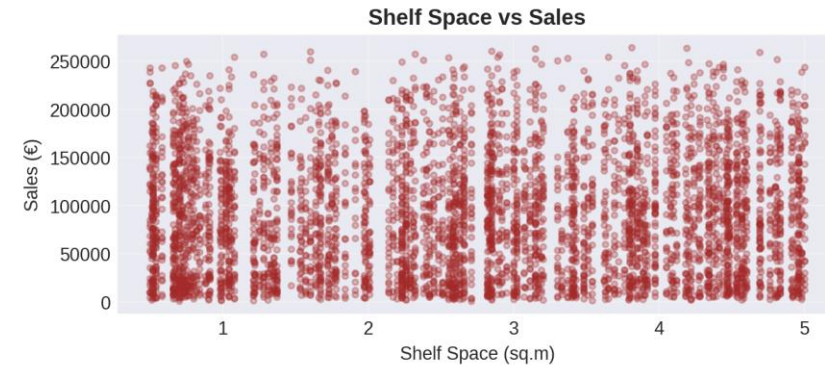
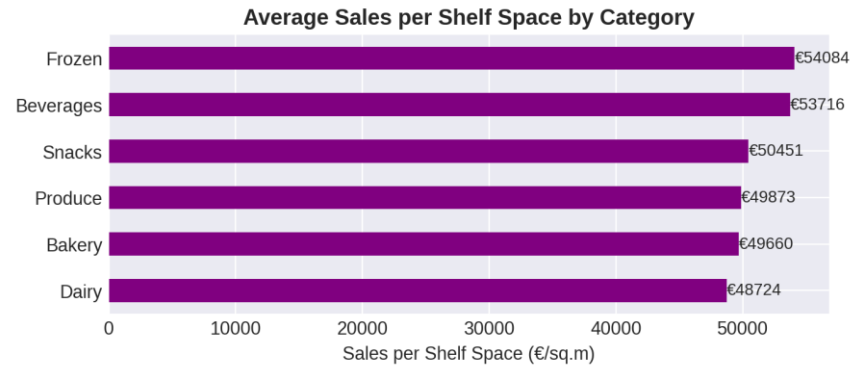
FINDING

Turnover (units sold per SKU) has a median of 64.2 with a long right tail up to 529. The slowest 1,250 products (≤ 43.9) generate €74.5M sales; the fastest 1,250 (≥ 110.0) generate €133.3M, 1.8× more.

ANALYTICAL INSIGHT

Slow movers carry 76 SKUs each on average against 24 for fast movers. Together they hold 35% of all SKUs and 25% of shelf space while earning €22.2k per m² vs €39.2k. Cutting SKU depth on slow movers is the fastest way to release working capital.

Shelf space bears no relation to what a product sells



FINDING

Category productivity is similar (€48.7k/m² Dairy to €54.1k/m² Frozen). At product level, space and sales barely correlate (0.03). The least productive quartile holds 30.7% of space and loses €42.8M; the top quartile holds 12.2% of space yet delivers 39% of sales.

ANALYTICAL INSIGHT

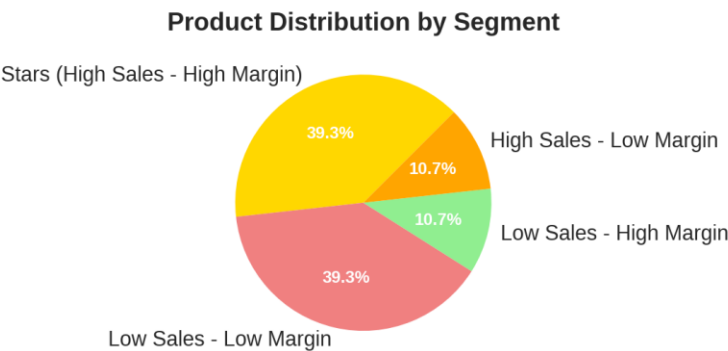
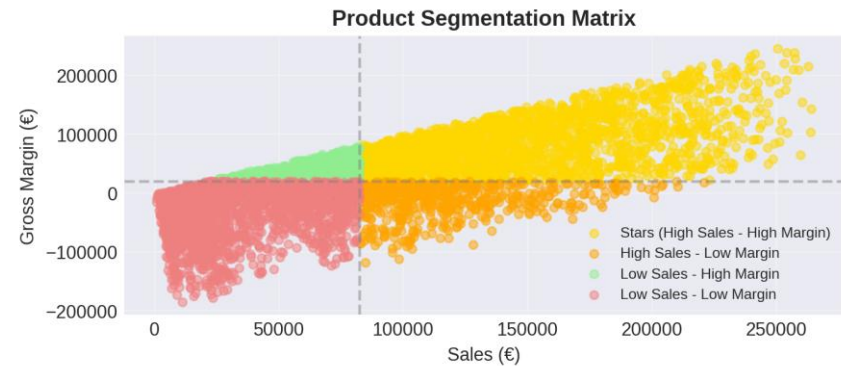
Space is allocated by habit, not performance. Top-quartile products are cramped (1.3 m² average) and likely to run out of stock; bottom-quartile products are over-spaced (3.4 m²). Moving space from the bottom to the top quartile will raise sales per m² without adding any store space.

SECTION 04

Profitability analysis

Which products deserve priority space, and which have negative or low margins.

Stars carry the business; the low-low quadrant loses €69M



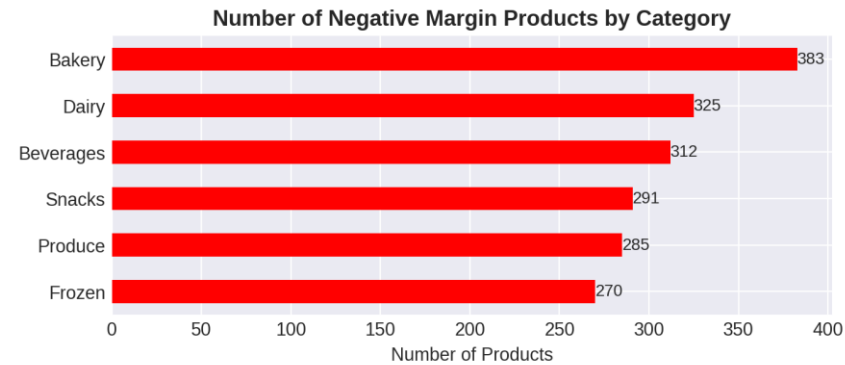
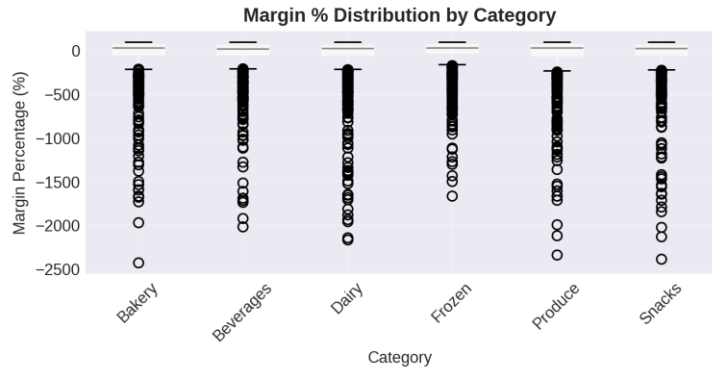
FINDING

Split at median sales (€82.4k) and margin (€20.4k): Stars are 39.3% of products, 64.7% of sales and €170.0M margin. Low sales–low margin is also 39.3% but loses €69.0M. High sales–low margin (10.7%) loses €7.7M; low sales–high margin (10.7%) earns €21.5M.

ANALYTICAL INSIGHT

Stars should get priority space and stock. The low-low quadrant is the core elimination pool. High sales–low margin items have proven demand, so fix price or cost before delisting. Low sales–high margin items are growth candidates for better placement.

Every category has a long tail of deep loss-makers



FINDING

The median margin rate is 25.2%, but 18.6% of products lose more than 100% of their sales value, down to -2,430%. Negative-margin counts track category size: Bakery 383, Dairy 325, Beverages 312, Snacks 291, Produce 285, Frozen 270. 2,058 products earn 40%+ margin.

ANALYTICAL INSIGHT

The notebook's average margin of -63.4% is dragged down by these outliers; the true blended margin is 25.2%. Loss-makers make up 34–39% of every category, so the fix is a product-level cut applied across all categories, with Beverages (39.3%) and Dairy (38.8%) first.

SECTION 05

Recommendations

What to retain, replace or eliminate, and the revised assortment strategy.

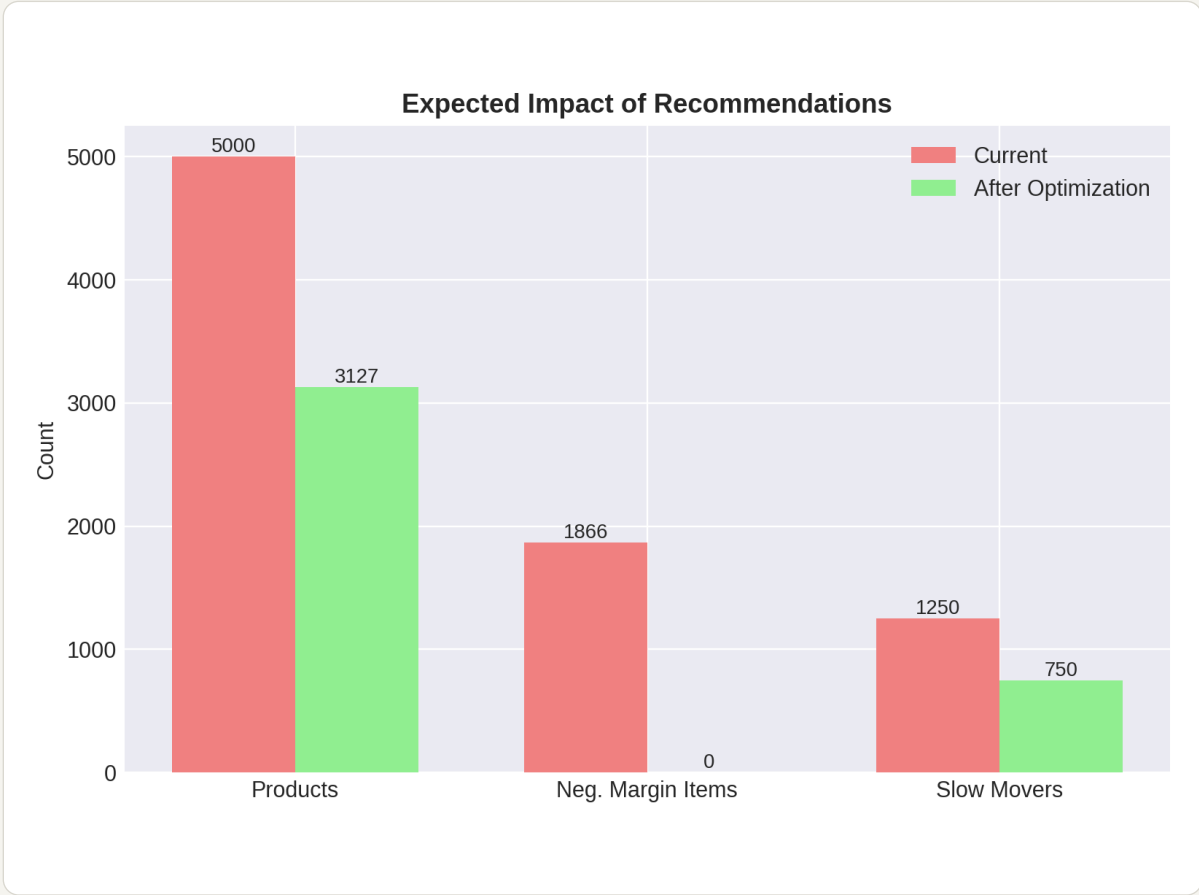
RECOMMENDATIONS · RETAIN, REPLACE, ELIMINATE

Every product gets one clear decision

Decision	Products	Sales share	Margin	Shelf share	Rule and action
Expand	1,186	42.9%	€121.0M	20.6%	Stars, top-quartile sellers and shelf-efficient items: more facings and deeper stock
Grow	298	3.3%	€9.4M	5.9%	Rated 4.0+, below-median sales, 20%+ margin: better placement and promotion
Retain	1,643	34.9%	€67.4M	36.0%	Profitable core: keep and review each season
Replace / reprice	220	2.3%	–€10.3M	4.4%	Loss-makers rated 4.0+ and bought often: renegotiate cost, reprice or swap for private label
Eliminate	1,653	16.6%	–€72.8M	33.2%	Negative margin, or low sales with low ratings, or very slow with loss on shelf

Built on the criteria: its 1,873 elimination candidates are split into 1,653 to delist and 220 to protect; its 1,318 focus products lose the 132 that also sit on the elimination list. Each product is counted once.

The range shrinks by 37% while margin more than doubles



FINDING

Removing the 1,873 flagged products takes the range from 5,000 to 3,127, clears all 1,866 negative-margin items and recovers €83.1M. The margin rate on the remaining €368.9M of sales rises from 25.2% to 53.6%. 5,115 m² of shelf is freed (37.5% of total).

ANALYTICAL INSIGHT

Sales at risk are €85.8M (18.9%), but part of that will transfer to the substitutes that stay. Keeping the 220 well-loved loss-makers on a reprice track limits lost shoppers. The slow-mover reduction (1,250 → 750) is an estimate and should be tracked after launch.

Every category needs pruning; the emphasis differs

Category	Products	Sales	Eliminate	Focus	Star	Priority move
Bakery	1,026	€91.6M	383	257	4101	Largest range: trim duplicate SKUs across brands
Dairy	838	€74.6M	327	209	3237	Low margin per product: cut losses, renegotiate
Beverages	793	€70.9M	315	198	4673	Lowest margin rate (21.7%): price and cost review
Snacks	768	€70.0M	292	198	3950	Highest rating (3.80): promote high-rated items
Produce	783	€71.7M	286	230	1608	Strong margin: shift space to top sellers
Frozen	792	€75.8M	270	226	4966	Best performer: expand space and range depth

Eliminate and focus counts are the category-level output. Star = highest-margin product in the category.

A phased plan from stopping losses to growing the core

IMMEDIATE · 0–1 MONTH

Stop the losses

- Delist the 1,653 eliminate SKUs, starting with the deepest loss-makers
- Move freed shelf space to the 1,186 expand products
- Raise stock depth on top-rated, high-frequency lines

SHORT TERM · 3 MONTHS

Fix and reposition

- Reprice or renegotiate the 220 protected loss-makers, and Brand D terms
- Promote the 298 grow products with better placement
- Reset planograms to sales per m²

LONG TERM · 6–12 MONTHS

Build the core

- Private label alternatives in high-margin categories
- Dynamic pricing for seasonal and perishable items
- Loyalty offers built around high-margin products
- Quarterly range review using this scorecard

APPENDIX

Definitions and data notes

DEFINITIONS

Margin rate = gross margin ÷ sales. Stock turnover = units sold ÷ SKU count. Shelf productivity = sales ÷ shelf space (m²). Segments split at median sales (€82.4k) and median margin (€20.4k).

DATA NOTES

37.3% of rows have unit cost above price, which explains margins as low as -2,430%; these costs should be checked before delisting. The blended margin rate (25.2%) is used as the baseline, not the average of ratios (-63.4%).